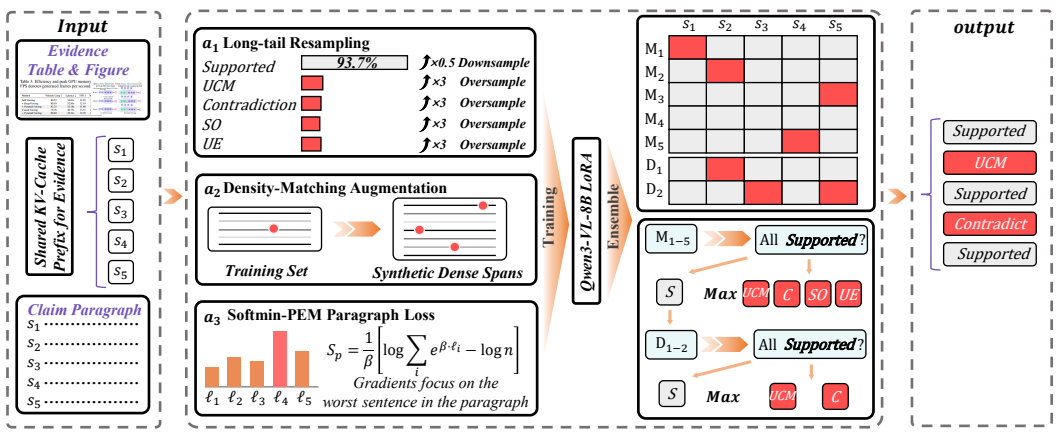

Input
Evidence
Table & Figure
Shared KV-Cache
Prefix for Evidence
s_1
s_2
s_3
s_4
s_5
Claim Paragraph
s_1
s_2
s_3
s_4
s_5
a_1 Long-tail Resampling
Supported
93.7%
×0.5 Downsample
UCM
×3 Oversample
Contradiction
×3 Oversample
SO
×3 Oversample
UE
×3 Oversample
a_2 Density-Matching Augmentation
Training Set
Synthetic Dense Spans
a_3 Softmin-PEM Paragraph Loss
$S_p = \frac{1}{\beta}\left[\log \sum_i e^{\beta \cdot \ell_i} - \log n\right]$
ℓ_1 ℓ_2 ℓ_3 ℓ_4 ℓ_5
Gradients focus on the worst sentence in the paragraph
Training
Qwen3-VL-8B LoRA Ensemble
s_1 s_2 s_3 s_4 s_5
M_1
M_2
M_3
M_4
M_5
D_1
D_2
M_{1-5}
All Supported?
S
Max
UCM
C
SO
UE
D_{1-2}
All Supported?
S
Max
UCM
C
output
Supported
UCM
Supported
Contradict
Supported